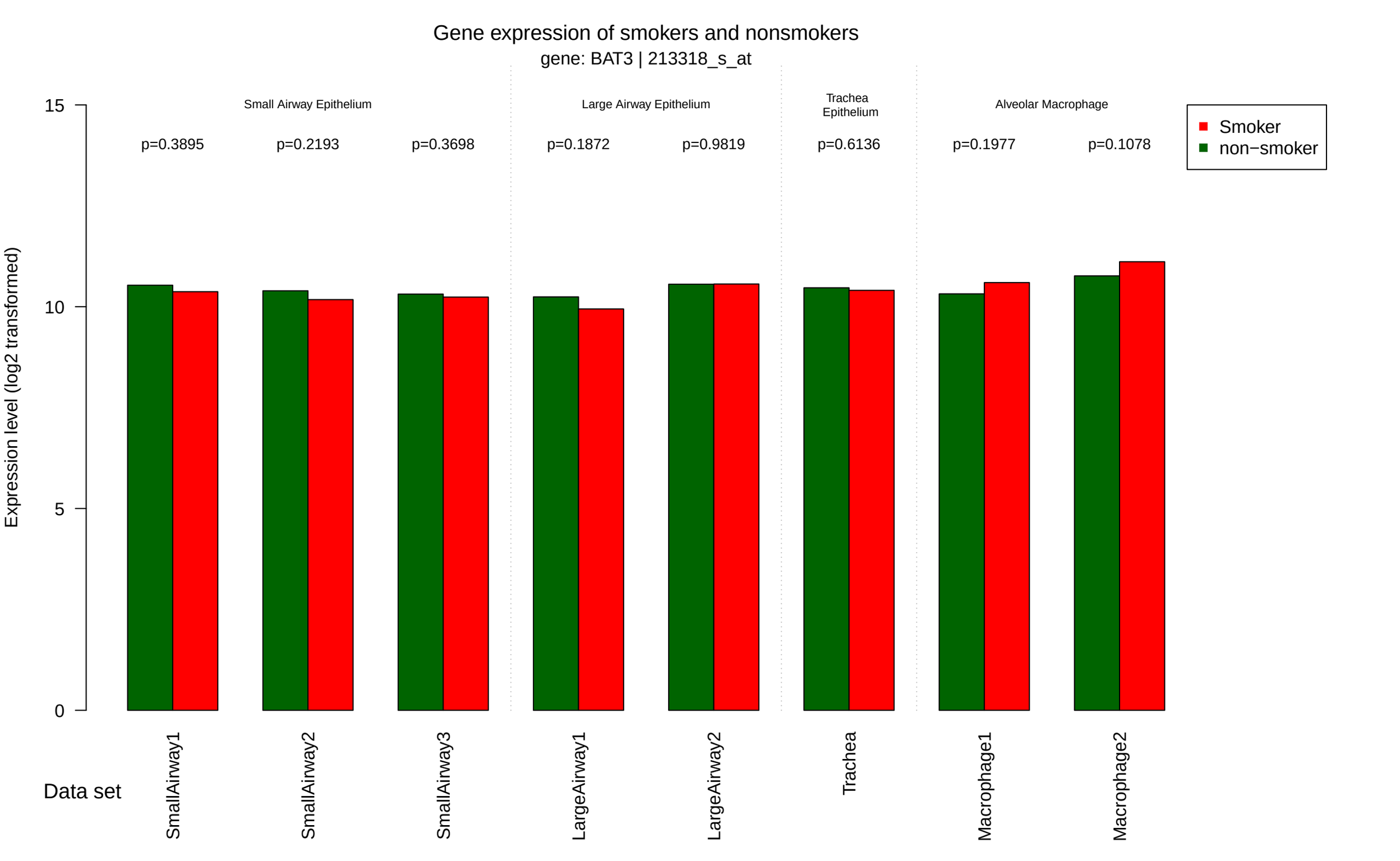
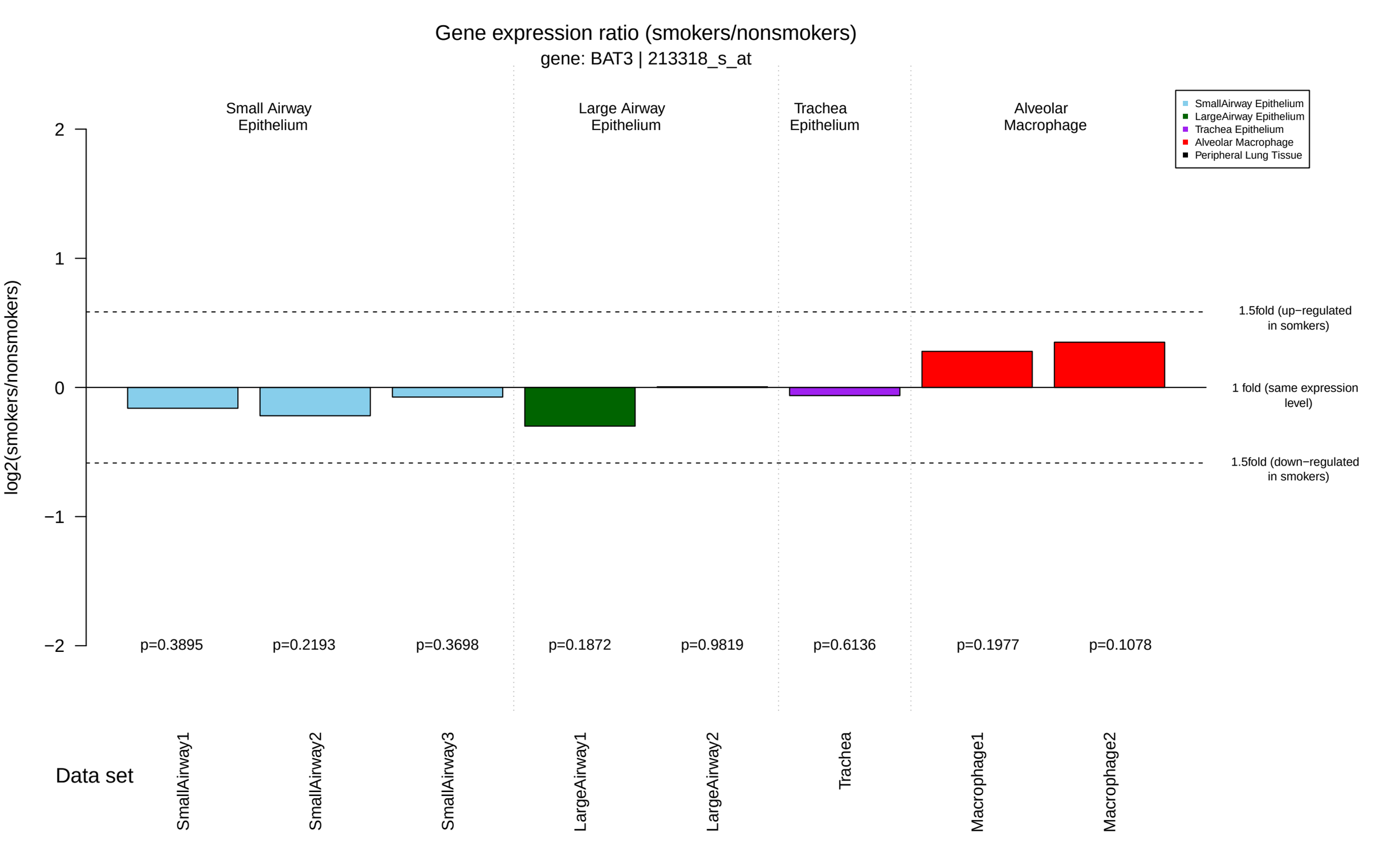
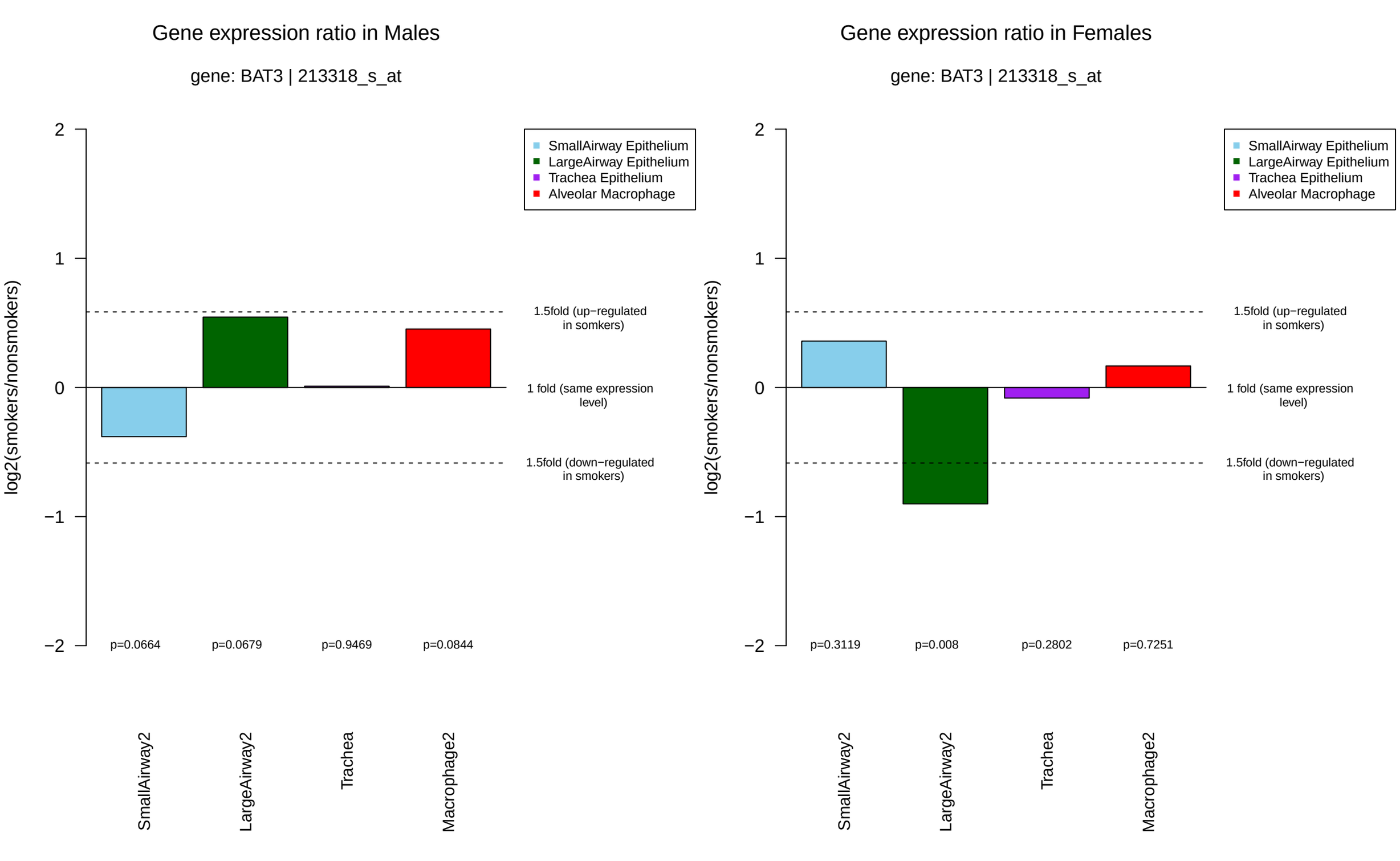
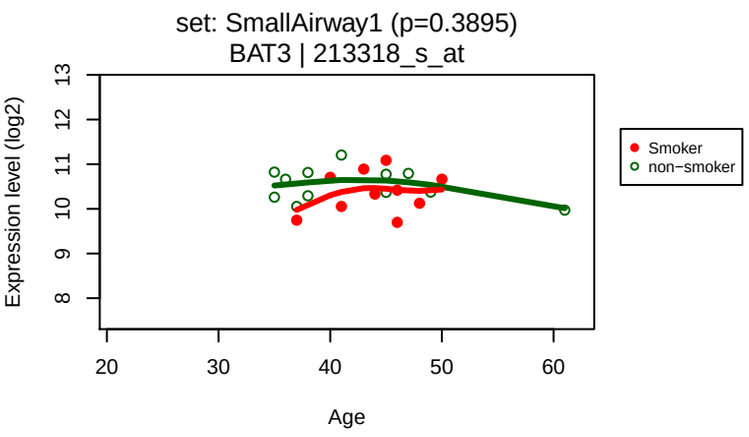
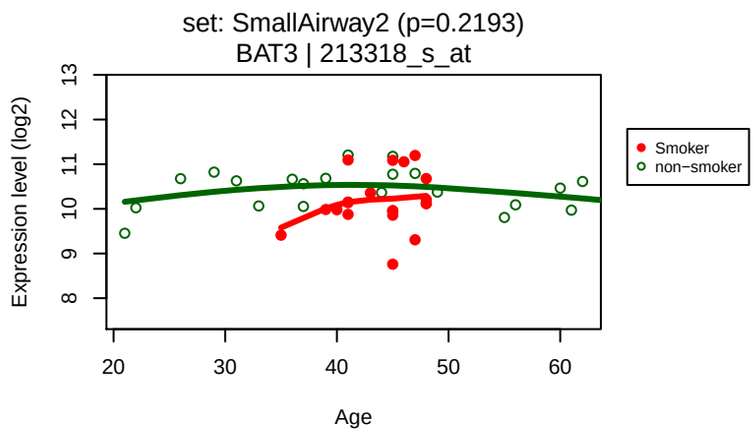
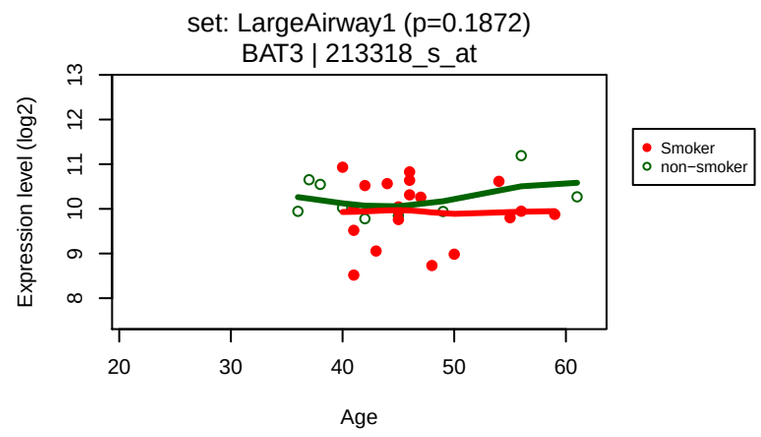

Gene expression of smokers and nonsmokers
gene: BAT3 | 213318_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3895
p=0.2193
p=0.3698
p=0.1872
p=0.9819
p=0.6136
p=0.1977
p=0.1078
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

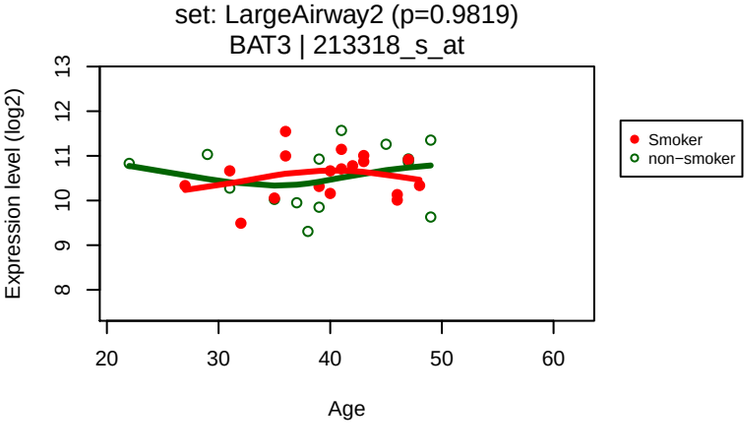

Gene expression ratio (smokers/nonsmokers)
gene: BAT3 | 213318_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3895
p=0.2193
p=0.3698
p=0.1872
p=0.9819
p=0.6136
p=0.1977
p=0.1078
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

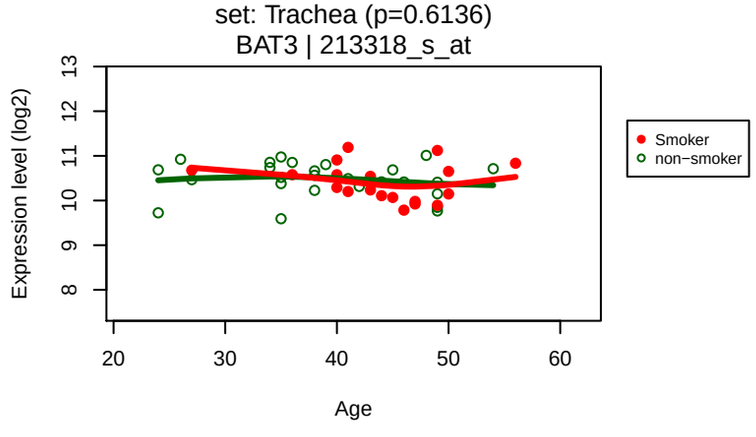

Gene expression ratio in Males
gene: BAT3 | 213318_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0664
p=0.0679
p=0.9469
p=0.0844
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BAT3 | 213318_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3119
p=0.008
p=0.2802
p=0.7251
SmallAirway2
LargeAirway2
Trachea
Macrophage2

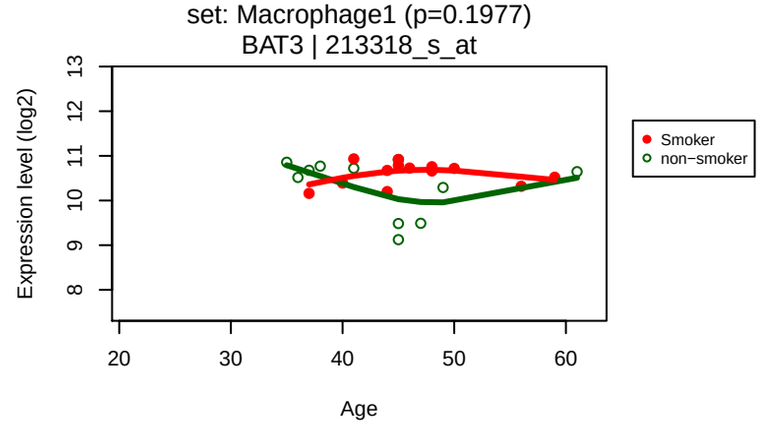

set: SmallAirway1 (p=0.3895)
BAT3 | 213318_s_at
Expression level (log2)
Age
Smoker
non−smoker

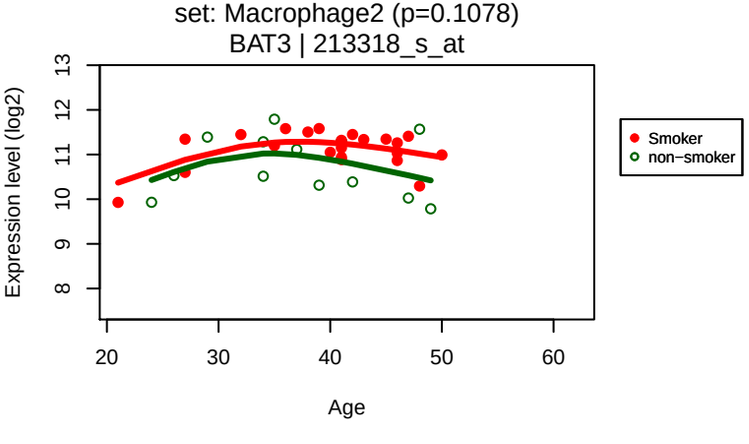

set: SmallAirway2 (p=0.2193)
BAT3 | 213318_s_at
Expression level (log2)
Age
Smoker
non−smoker

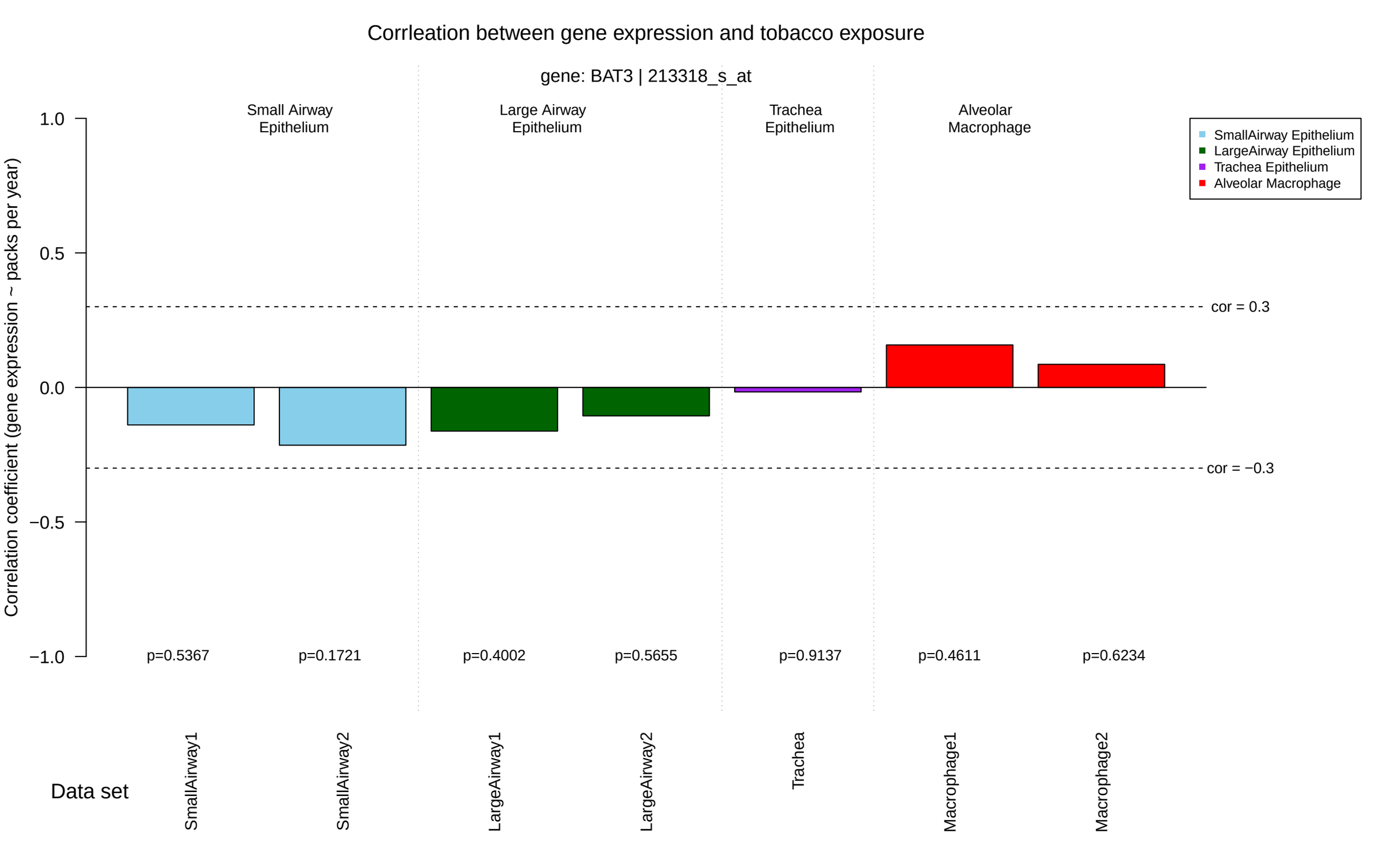

set: LargeAirway1 (p=0.1872)
BAT3 | 213318_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9819)
BAT3 | 213318_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6136)
BAT3 | 213318_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1977)
BAT3 | 213318_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1078)
BAT3 | 213318_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BAT3 | 213318_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5367
p=0.1721
p=0.4002
p=0.5655
p=0.9137
p=0.4611
p=0.6234
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2